Seeing Through the Blackout: Gesture-Based Theatrical Lighting Control Under Adversarial Illumination

Amogh Bajpai
Independent Researcher
Noida, India
hi@amoghbajpai.com

***Abstract*—A theatrical lighting cue is conventionally called by a stage manager and executed by a board operator. Productions without the staffing to fill those roles leave the expressive range of the rig unused. We present *Lights*, a markerless, CPU-only, browser-resident system that maps five hand gestures onto five lighting cues and emits a real 512-channel DMX512 universe rather than a screen animation. The vocabulary is structured so that its confusion matrix is close to block-diagonal by construction, which permits a purely rule-based implementation with every threshold expressed in scale-invariant hand-span units and no calibration step. We argue that accuracy is the wrong headline metric for this domain, because the cost of a false positive — a stage blacked out mid-scene, in front of an audience — is categorically greater than the cost of a false negative, and we adopt false triggers per hour in its place. We report three findings that emerged only from continuous operation and that per-gesture evaluation is structurally unable to observe: landmark tracking fails at the completion of self-occluding gestures; two individually correct detectors can form an interference cascade in which one cue fires on another's motion signature and then suppresses it; and cues in operator contexts must be performable without visual confirmation. We further show that end-to-end latency is dominated by temporal evidence accumulation and fixture rise time rather than by inference, which redirects optimisation away from model speed and toward early classification. Finally we name *adversarial illumination* — the rig corrupting the sensing path through which it is commanded — as the constraint that distinguishes this domain from every adjacent one, and propose near-infrared spectral separation as its resolution. The system is deployed and self-instrumenting; the evaluation protocol is specified in full but has not yet been executed.**

***Index Terms*—Gesture Recognition, Theatrical Lighting, DMX512, Human-Computer Interaction, Hand Tracking, MediaPipe, False Triggers, Live Performance, Adversarial Illumination**

I. Introduction

A theatrical lighting cue is conventionally the product of two people. A stage manager calls the cue; a board operator executes it on a lighting desk. The division of labour is long established, robust, and rehearsed — and it presumes a production with the personnel to staff it. [1]

Student theatre, community productions, dance recitals and fringe work routinely have neither role filled. The lighting in these venues is either run by an unrehearsed volunteer reading from a paper cue sheet, or abandoned altogether in favour of a static wash held for the duration of the performance. The expressive range of the rig, which is considerable even in a modest venue, goes unused because there is nobody free to operate it.

This paper asks a narrow question: can a lighting cue stack be driven directly by the hand, markerless, on hardware already present in the room, reliably enough to place in front of a paying audience? We present *Lights*, a browser-resident, CPU-only system that maps five hand gestures onto five lighting cues and emits a real 512-channel DMX512 universe rather than a screen animation, an overview of which is shown in Figure 1.

The phrase carrying the weight in that question is *reliably enough*, and we argue it is not measured by accuracy. A gesture interface for a text editor that misfires once an hour is mildly irritating. A lighting system that misfires once an hour will, at some point during a run, black out a stage mid-scene. The failure is neither graceful nor recoverable, and it is witnessed by an audience. We therefore adopt **false triggers per hour** (FTR) under ordinary, non-cueing movement as the headline metric, and treat per-gesture accuracy as a secondary diagnostic. [2]

The contributions of this work are as follows. We present a five-cue gesture vocabulary engineered so that its confusion matrix is close to block-diagonal by construction, before any classifier is introduced. We describe a rule-based reference implementation with all thresholds expressed in scale-invariant hand-span units, deployed and self-instrumenting. We report three findings that emerged only from building and operating the system, none of which is visible to per-gesture evaluation. We decompose end-to-end latency and show inference to be a minority term, which redirects the optimisation target away from faster hardware. Finally, we name *adversarial illumination* as the constraint that distinguishes this domain from every adjacent one, and propose a spectral resolution.

We are explicit at the outset about what this paper does not contain. The system has been built, deployed and operated by a single user. It has not been evaluated with participants, and no FTR figure is reported. The protocol the instrumentation was built to serve is specified in full; executing it is future work.

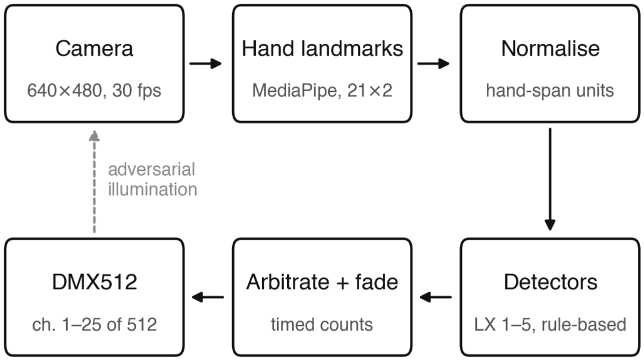

Fig. 1. System overview: camera to hand landmarks to five rule-based detectors to a 512-channel DMX universe. The dashed return path is the constraint that characterises the domain — the rig the system commands illuminates the scene the system must observe.

II. Problem Statement

Two problems define this domain, and the second appears not to have been articulated in the literature.

The first is the staffing problem described above. Cue execution requires a dedicated operator whose entire attention is committed to the desk for the duration of the performance. In productions without that person, the rig is underused. Any solution must run on commodity hardware, require no installation, no calibration step, and no worn markers, because the deployment context is one in which none of those will happen.

The second problem is structural, and it is a closed loop: **the system's own output is an adversary to its own input**. A blackout — the single most important cue in the vocabulary — removes the illumination the camera depends upon, and the system must remain controllable in the state it has just created. A saturated colour wash shifts skin tone far outside the distribution on which a hand tracker was trained. Atmospheric haze, standard theatrical practice for beam visibility, scatters the image. In every case the disturbance is produced by the artistic intent of the production, and is therefore not something the system may ask to have reduced.

The control channel and the artistic channel occupy the same physical medium, and the artistic channel is under the control of the very system attempting to observe through it. We call this *adversarial illumination*. It is, we argue, the constraint that separates stage lighting from every adjacent gesture-control domain, and it admits a principled resolution which we set out in Section VI.

III. Related Work

Gesture control of lighting exists, but not for this application. Work on gesture and speech control of *surgical* theatre lighting addresses the sterile field problem: a surgeon cannot touch a control surface without breaking asepsis. [3] The domain shares a keyword with ours and very little else. The task is aiming a task light, the environment is uniformly and brightly lit by design, there is no cue structure, no audience, and no consequence to a misfire beyond inconvenience.

TABLE I
The five-cue vocabulary

ID	Cue	Gesture	Kind	Hands
LX 1	Blackout	Clap twice	dynamic	2
LX 2	General cover	Both palms open, raised	static	2
LX 3	Centre special	Point, hold	static	1
LX 4	Master dim	Open hand, move up/down	continuous	1
LX 5	Colour state	Swipe sideways	dynamic	1

Architectural and smart-building lighting control targets ambience and occupancy response over long time constants. [4] Stage lighting inverts nearly every requirement of that setting: cues are discrete, timed to the second, rehearsed in advance, and dramatic rather than functional.

Within theatrical lighting proper, prior work has addressed interaction paradigms without addressing gesture. A contribution towards intuitive theatrical lighting control critiques the conventional desk workflow and proposes alternatives, but the interaction modality throughout remains a control surface. [5] To our knowledge, the specific cell — markerless gesture recognition for theatrical cue calling — is unoccupied.

Commercial performer tracking is mature and deployed at professional scale. BlackTrax, zactrack and Robe RoboSpot are all in regular use in large venues. [6]–[8] They are, however, solving a different problem. All are marker- or beacon-based, requiring the performer to carry a tracked device, and all track *position* in order to aim a followspot or moving head at a body. The present work tracks *intent* in order to call a cue: markerless, no worn hardware, and at a small fraction of the cost. Position tracking answers *where is the performer*; cue calling answers *what should the rig do now*. The two are orthogonal, and conflating them is the most common misreading of this work.

The core reliability problem is a specific case of the *Midas touch* problem, first characterised by Jacob in the context of eye-tracking interfaces: when an always-on channel is used for input, the system cannot distinguish deliberate command from incidental activity. [2] Our contribution here is not the observation but the metric — reframing it for a domain in which the cost of a false positive is categorically greater than the cost of a false negative, and instrumenting the system accordingly.

We build on MediaPipe hand landmark estimation as an unmodified upstream component. [9] The contribution of this work lies downstream of the landmarks: the vocabulary, the detectors, the arbitration between them, the fade engine, and the instrumentation.

IV. Methodology

A. Cue Vocabulary Design

The vocabulary is deliberately small and deliberately structured. It comprises exactly five cues, given in full in Table I. Facial expression detection is explicitly out of scope.

The structure *is* the design. The five cues split two static, two dynamic and one continuous; and two two-handed against three one-handed. These axes are close to independent, which separates most pairs of cues before any classifier is asked to distinguish them. Static and dynamic cues are separated by motion energy across the temporal window, a cheap and robust scalar. Two-handed and one-handed cues cannot collide at all, because hand count gates the detector branch entirely. The continuous cue is structurally distinct in that it produces a value rather than an event, and so is never in competition for a discrete firing.

The intent is that the confusion matrix is close to block-diagonal by construction. This is a design-time substitute for discriminative training, and it is what permits a purely rule-based implementation to function at all.

The residual risk is deliberately concentrated in one place. LX 5, the lateral swipe, carries the highest false-trigger risk in the set: a sideways hand movement is the single most common incidental motion a person makes, and unlike the other four cues it has no hold, no count and no two-hand requirement to protect it. Section VI reports this predicted risk materialising in an entirely unanticipated form.

B. Fade Semantics

Discrete cues execute on a timed fade, as they would on a lighting desk: the blackout runs 1.0 s out and 1.8 s back in, the general cover 3.0 s, the centre special 2.5 s, and the colour state 2.0 s.

The master dim deliberately does *not* fade; it tracks the hand directly. A fade on the master would be experienced as lag rather than as craft, because the operator's hand *is* the control surface and a control surface that lags is broken. This distinction — between a cue that is *called* and a parameter that is *held* — is not incidental. It is the reason the vocabulary requires a continuous class at all.

C. System Architecture

The pipeline proceeds from camera, to MediaPipe hand landmark estimation, to normalisation, to five rule-based detectors, to arbitration, to a fade engine, and finally to a 512-channel DMX universe.

Every geometric quantity is divided by the hand span — the wrist to middle-finger MCP distance — so that all thresholds are expressed in hand-spans rather than pixels or normalised frame units. A hand near the camera and a hand across the room therefore produce identical numbers. This removes any need for per-operator or per-room calibration, which matters precisely because the deployment context is one in which nobody will perform a calibration step.

The rig comprises seven fixtures patched across DMX channels 1 to 25 within a standard 512-channel universe. Values are generated exactly as they would be transmitted, and the browser renders the same bytes it would send. Substituting an sACN socket for the on-screen renderer drives physical fixtures with no change to the control path. This matters for the central claim of the system: it emits lighting control data, not an animation of lighting.

TABLE II
PRINCIPAL DETECTOR PARAMETERS

Cue	Parameter	Value
LX 1	Contact separation	1.3 spans
LX 1	Re-arm separation	2.2 spans
LX 1	Separation at track loss	2.4 spans
LX 1	Closing speed	4.5 spans/s
LX 1	Grace after pair lost	260 ms
LX 1	Re-arm timeout	240 ms
LX 1	Inter-clap gap	120–900 ms
LX 2	Hold	400 ms
LX 2	Wrist height	0.62
LX 3	Hold	500 ms
LX 3	Stillness per frame	0.035
LX 4	Minimum extended fingers	3
LX 4	Smoothing coefficient	0.4
LX 5	Window	350 ms
LX 5	Travel	0.3 frame widths
LX 5	Minimum samples	4 frames
LX 5	Cooldown	1500 ms
LX 5	Two-hand lockout	800 ms
—	Global cooldown	350 ms
—	Hand-count debounce	110 ms

The prototype is a static browser application requiring no installation and running CPU-only. The MediaPipe WASM runtime and the hand-landmarker model are vendored into the application bundle rather than fetched from a content delivery network, on the principle that a live performance must not depend on a third-party host being reachable.

D. Detector Design

All five detectors are rule-based. **No model is trained**, and this is a deliberate methodological choice rather than a limitation of effort. Rules establish that all five cues are achievable, and they provide the baseline that a learned classifier must beat — which is the comparison worth reporting. What rules cannot supply is a calibrated *that was not a cue* score, and it is precisely this gap that motivates a null class and a small temporal model as the next step, driven by measured FTR rather than by preference.

Table II gives the principal parameters. All are declared in a single location, in hand-spans or milliseconds.

Two of these values are not tuning constants but direct consequences of the findings reported in Section VI: the LX 1 grace-after-loss window, and the LX 5 two-hand lockout.

V. Experimentation and Results

The prototype was instrumented for measurement rather than for demonstration. It reports per-stage latency, logs every cue firing with a timestamp, permits any firing to be flagged as unintended, computes FTR live over session time, and exports a session as CSV. It additionally exposes a per-cue condition readout showing which condition a cue is waiting on, with the measured value beside the threshold it missed — because a gesture system cannot be debugged through a console by someone who is holding a pose.

The evaluation protocol these facilities exist to serve has not yet been run, and this section reports no measured results from it. The system's present status is a working prototype, operated by a single user. We state the protocol in full below so that the instrumentation described above is legible as design rather than as decoration.

A. Experimental Setup

The planned evaluation proceeds in four stages:

1. **Armed idle recording.** A two-hour continuous session with the system armed and the operator engaged in ordinary non-cueing activity, yielding the headline FTR figure.
2. **Participant study.** Multiple operators, per-cue F1, and cross-user generalisation. The prototype has to date been operated by its author alone, which is the most severe limitation of the present work.
3. **Rules versus learned comparison.** A null-class temporal model (GRU or TCN over 30-frame windows) evaluated against the rule baseline on identical recorded sessions, compared on FTR rather than on accuracy.
4. **Physical deployment.** Real fixtures, real haze and a production context, in order to test the adversarial illumination claim directly.

B. Latency Analysis

One result is available from the current instrumentation, and it is structural rather than empirical in character. Inference accounts for approximately 8% of the end-to-end budget; the dominant terms lie elsewhere, and the decomposition is given in Figure 2.

The first is **temporal evidence accumulation**. A classifier cannot commit until it has observed enough of a gesture to be confident. For the dynamic cues this is the 350 ms swipe window or the inter-clap interval; for the static cues it is the hold, 400 to 500 ms. This is an epistemic cost, not a computational one. The second is **fixture rise time**: physical fixtures have electromechanical and thermal response characteristics, and a tungsten fixture cannot change state instantaneously however quickly it is commanded.

The consequence is that faster hardware does not meaningfully improve this system. Reducing inference cost to zero would recover under a tenth of the observed latency. The available lever is instead **early or anticipatory classification** — committing to a cue from a partial sequence, before the gesture has completed, and accepting the resulting precision-latency trade-off explicitly. This is a substantively different research direction from the model-efficiency work that a naive reading of the latency requirement would suggest.

We note a measurement boundary. Camera exposure and USB transport sit upstream of anything observable from within a browser. The instrumentation reports frame interval, landmark inference and cue classification separately and states this boundary explicitly, rather than quoting a flattering end-to-end figure it cannot substantiate. A true end-to-end measurement requires an external clock and a photodiode.

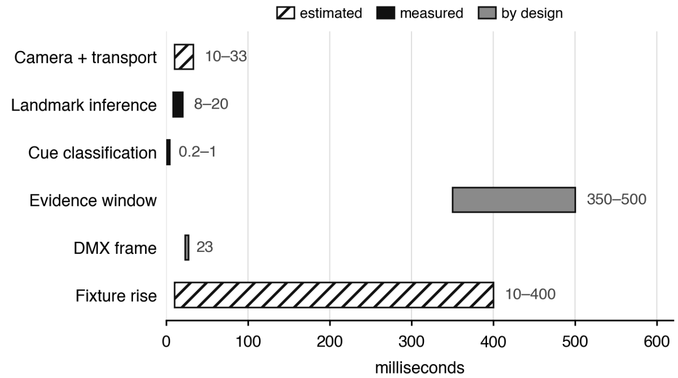

Fig. 2. End-to-end latency budget. Inference is a minority term; evidence accumulation and fixture rise time dominate. Bars are ranges, not point estimates, and are distinguished by provenance: *measured* in the browser instrumentation, *by design* from the detector thresholds and the DMX refresh rate, and *estimated* for the two terms that lie outside what a browser can observe.

VI. Challenges and Solutions

The following emerged from building and operating the system. We report them as findings rather than as engineering notes, because each generalises beyond this implementation and because none is visible to the per-gesture accuracy evaluation by which this class of system is normally assessed.

A. The Tracker Goes Blind as the Gesture Completes

A clap cannot be detected from contact, because contact is the moment the tracker fails. When two palms meet, the merged region ceases to resemble two hands and the landmark estimator drops the pair. The frame in which the gesture is completed is precisely the frame in which observation is lost.

The detector is therefore built to infer the event from its shoulders rather than its peak: fast closing motion at 4.5 spans per second, followed by loss of the pair while separation was still below 2.4 spans, within a 260 ms grace window. The event is reconstructed from the approach and the disappearance rather than observed directly.

We suggest this generalises. Any gesture whose defining moment is self-occluding — contact, grasp, fist closure, hand over hand — will be invisible at its own climax to a landmark-based tracker. Detectors for such gestures must be designed to read the envelope, not the event.

B. Cross-Cue Interference Cascade

This is the most consequential finding, and the one with the clearest methodological implication.

A clap throws both hands laterally through the frame at speed. At the instant the pair merges, the tracker reports a single hand travelling fast — which is an exact match for the signature of a lateral swipe. Every clap therefore fired a spurious colour cue.

The failure then compounded. The spurious colour cue held the global cooldown open, and that cooldown suppressed the second clap — the very gesture that had caused it. The blackout became unreliable, and the proximate cause was a detector for an entirely different cue firing on the blackout's own motion signature.

Neither detector was wrong in isolation. Tested individually, each behaved correctly against its own gesture. The failure exists only in the interaction between them, and only under continuous operation.

The resolution was structural rather than parametric: two separate motion histories, so that two-handed motion never reaches the swipe detector, in combination with an 800 ms two-hand lockout on LX 5. A swipe is one-handed by definition, so anything within that window of seeing two hands cannot be one.

The methodological point is the important one. Per-gesture accuracy testing is structurally incapable of detecting this failure mode. Presenting gesture i and recording whether cue i fires will report every detector as correct. The cascade appears only under continuous operation with a false-trigger log. This is a second and independent argument for FTR as the primary metric, additional to the cost-asymmetry argument given in the introduction.

C. Cues Must Be Performable Without Looking

The master dim originally required a flat palm facing downward, with palm orientation tested explicitly. It worked in isolation and failed in use.

The reason is that an operator watches the stage, not their own hand. A gesture requiring a specific hand orientation requires visual confirmation in order to be performed reliably, and visual attention is the one resource an operator running a show does not have. Orientation testing was removed entirely; any open hand with at least three extended fingers now drives the master.

The resulting design rule is that, in an operator context, a cue which must be looked at in order to be performed is not a usable cue. This eliminates a substantial region of the otherwise attractive gesture design space, notably most precision poses and anything depending on finger-level configuration.

D. Adversarial Illumination

The closed-loop problem set out in Section II admits a principled resolution: spectral separation. Near-infrared imaging with infrared illuminators places the control channel outside the visible band entirely, so that the designer may do anything at all with visible light without touching the sensing path. Theatrical haze scatters less at near-infrared wavelengths, and infrared illumination is invisible to an audience.

We state plainly that this is argued and not yet demonstrated. The current prototype uses a visible-light webcam. Validating the closed-loop problem requires real fixtures illuminating a real lens, and this is the single most important item of future work: without light on a lens there is no evidence for the problem the system is ultimately about.

VII. Conclusion

We have presented a markerless, CPU-only, browser-resident system that maps five hand gestures onto five theatrical lighting cues and emits real DMX512. We have argued that false triggers per hour, rather than accuracy, is the appropriate headline metric for interfaces whose failure cost is asymmetric, and we have shown a concrete failure — the cross-cue interference cascade — that per-gesture accuracy testing is structurally unable to observe.

We have reported that landmark tracking fails at the completion of self-occluding gestures, requiring detectors that read the envelope rather than the event; that cues in operator contexts must be performable without visual confirmation; and that latency in this domain is epistemic and electromechanical rather than computational, which redirects the optimisation target from model speed toward early classification. We have further named adversarial illumination as the constraint that distinguishes this domain, and proposed near-infrared spectral separation as its resolution.

The system augments the operator; it does not replace one. A human veto is retained at all times, and we regard this as a design commitment rather than a transitional safeguard.

The limitations are material and we do not minimise them. The system has been operated by a single user, so no claim of cross-user generalisation is supported. No FTR figure is reported. The adversarial illumination argument is motivated but not empirically demonstrated. The thresholds in Table II were set by iterative observation by one operator and should be regarded as a starting configuration rather than a tuned one. Finally, the vocabulary comprises five cues, whereas a production cue stack is routinely an order of magnitude larger; whether the block-diagonal design property survives that scaling is untested, and we suspect it is the binding constraint on this approach.

References

[1] R. Pilbrow, *Stage Lighting Design: The Art, The Craft, The Life*. London: Nick Hern Books, 2008.

[2] R. J. K. Jacob, "What you look at is what you get: Eye movement-based interaction techniques," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '90)*. Seattle, WA, USA: ACM, 1990, pp. 11–18.

[3] N. Zargham, A. V. Reinschluessel, A. Mühlenbrock, T. Muender, T. Cetin, V. N. Uslar, D. Weyhe, R. Malaka, and T. Döring, "Using gesture and speech to control surgical lighting systems: Mixed methods study," *JMIR Human Factors*, vol. 12, no. 1, p. e70628, 2025.

[4] E. Juntunen, P. Myöhänen, J. Häikiö, E.-M. Sarjanoja, O. Tapaninen, and V. Pentikäinen, "Orchestrating light: Gesture-based control of smart lighting," *Lighting Research & Technology*, vol. 49, no. 8, pp. 943–965, 2017.

[5] D. Schwarz, S. Cunningham, and R. Woodford-Smith, "Towards a new paradigm for intuitive theatrical lighting control," in *Proceedings of EVA London 2019: Electronic Visualisation and the Arts*. London, UK: BCS Learning and Development Ltd., 2019, pp. 360–367.

[6] CAST Group of Companies, "BlackTrax real-time tracking for lighting," https://cast-soft.com/blacktrax/, accessed: 2026-08-09.

[7] zactrack Lighting Technologies GmbH, "zactrack automated follow system," https://www.zactrack.com/, accessed: 2026-08-09.

[8] Robe Lighting, "RoboSpot remote follow system," https://www.robe.cz/robospot, accessed: 2026-08-09.

[9] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang, and M. Grundmann, "Mediapipe hands: On-device real-time hand tracking," *arXiv preprint arXiv:2006.10214*, 2020.